Curriculum Vitae

Emidio Capriotti

Personal Information

Name: Emidio Capriotti
Nationality: Italian
Date of birth: February, 1973
Place of birth: Roma, Italy
Languages: Italian, English, Spanish

Positions

- **Oct 2019-present** Associate Professor: Department of Pharmacy and Biotechnology (FaBiT). University of Bologna, Bologna, Italy
- **Nov 2024-present** Joint Faculty, Computational Genomics Platform, IRCCS University Hospital of Bologna, Bologna.
- **2016-2019** Senior Assistant Professor (RTD type B): Department of Pharmacy and Biotechnology (FaBiT) and Department of Biological, Geological, and Environmental Sciences (BiGeA). University of Bologna, Bologna, Italy.
- **2015-2016** Junior Group Leader: Institute of Mathematical Modeling of Biological Systems, University of Düsseldorf, Düsseldorf, Germany
- **2012-2015** Assistant Professor: Division of Informatics, Department of Pathology, University of Alabama at Birmingham (UAB), Birmingham (AL), USA.
- **2011-2012** Marie-Curie IOF: Contracted Researcher at the Department of Mathematics and Computer Science, University of Balearic Islands (UIB), Palma de Mallorca, Spain.
- **2009-2011** Marie-Curie IOF: Postdoctoral Researcher at the Helix Group, Department of Bioengineering, Stanford University, Stanford (CA), USA.
- **2006-2009** Postdoctoral Researcher in the Structural Genomics Group at Department of Bioinformatics and Genetics, Prince Felipe Research Center (CIPF) Valencia, Spain.
- **2004-2006** Contract researcher at Department of Biology, University of Bologna, Bologna, Italy.
- **2001-2003** Ph.D student in Physical Sciences at University of Bologna, Bologna, Italy.

Education

- **Sep 2004** Master in Bioinformatics (first level)
  University of Bologna, Bologna (Italy).
- **Jun 2004** Ph.D. in Physical Sciences
  University of Bologna, Bologna (Italy).
- **Jul 1999** *Laurea* (B.S.) Degree in Physical Sciences, score 106/110
  University of Bologna, Bologna (Italy).

Research funding

- **2026-2028** Funding institution: ELIXIR Europe – Human Data and Translational Research Projects: ENIGMA and BioChef.
  Role: WP co-leader.

- **2023-2026** Funding institution: Italian Ministry of Health. PNRR 2022 – Rare Disease
  Project: Multiomic strategies to implement the diagnostic workflow of rare diseases (PNRR-MR1-2022-12376067).
  Role: Unit Coordinator. PI: Alfredo Brusco.
- **2022-2023** Funding institution: ELIXIR Europe – Implementation Studies
  Project: Commissioned Service on Rare Disease and Federated European Phenome-Genome Archive (EGA).
  Role: WP co-leader and participant.
- **2019-2023** Funding institution: Italian Ministry of Education Research and University (MIUR). – PRIN 2017
  Project: Integrative tools for defining the molecular basis of the diseases: computational and experimental methods for protein variant interpretation (PRIN-201744NR8S).
  Role: Unit Coordinator. PI: Piero Fariselli
- **2018-2020** Funding institution: National Institute of Health (USA)
  Project: Role of VpreB in immunoglobulin antigen binding site selection (1R21AI134027-01A1)
  Role: Consultant. PI: Harry Schroeder.
- **2016-2018** Funding institution: Spanish Ministry of Science and Innovation.
  Project: Bioinformatics applications in phylogenetics, metagenomics, systems biology and cancer genomics. (MEC: DPI2015-67082-P)
  Role: Investigator. PI: Francesc Andreu Rossello.
- **2015-2017** Funding institution: National Institute of Health (USA)
  Project: The pre-BCR CDR-H3 sensing site and H chain selection (1R21AI117703-01A1).
  Role: Co-Investigator. PI: Harry Schroeder.
- **2009-2012** Funding institution: European Union - Marie Curie International Outgoing Fellowships.
  Project: New methods to evaluate the impact of single point protein mutation on human health (PIOF-GA-2009-237225).
  Role: Beneficiary
- **2006-2008** Funding institution: European Union - Marie Curie Reintegration Grant - European Union.
  Project: RNA Comparative Modeling. (IRG39722)
  Role: Postdoc. PI: Marc A. Marti-Renom.
- **2004-2006** Funding institution: European Union - VI Framework Programme -
  Project: Biosapiens Network of Excellence, A European Virtual Institute for Genome Annotation.
  Role: Postdoc. PI: Janet Thornton.
- **2003-2004** Funding institution: Italian Ministry of Education Research and University (MIUR) -: FISR2002
  Project: Hydrolases from Thermophiles: Structure, Function and Homologous and Heterologous Expression.
  Role: Research fellow. PI: Rita Casadio.

Other grants and fellowships

- **Sep 2026 – Jul 2028** Project: RaDAr-IT – Patient-Centered Data Pathways for Rare Disease Equity in Argentina.
  Funding institution: University of Bologna - Global South Cooperation Project.
  Role: Coordinator
- **Sep 2017 – Sep 2018** Project: Study and development of computational methods for cancer genome interpretation.
  Funding institution: University of Bologna - International Cooperation Grant UNIBO-

UCSD.
Role: Coordinator

- **Dec 2017** Italian Ministry of Education, Universities and Research: Fondo per il finanziamento delle attività base di ricerca 2017.
- **Jun 2012 – Jul 2012** Project: Computational methods to predict the functional impact of protein variations on alpha-galactosidase and the efficacy of pharmacological chaperone therapy.
Funding institution: EMBO - Short Term Fellowship for visiting KU Leuven, Leuven (Belgium)
Role: Recipient
- **Jul 2008 – Aug 2008** Project: Development of a method for RNA structure prediction using MODELLER program.
Funding institution: Valencian Government (Spain) - Short-term travel fellowship supervised by Prof. Andrej Sali, Departments of Biopharmaceutical Sciences and Pharmaceutical Chemistry, University of California, San Francisco (UCSF).
Role: Beneficiary
- **Aug 2005 - Dec 2005** Project: Implementation of new software for protein structure prediction.
Funding institution: University of Bologna - Marco Polo short-term travel fellowship supervised by Prof. Jeffrey Skolnick, Center of Excellence in Bioinformatics University at Buffalo (NY)
Role: Beneficiary
- **Sep 2001 - Sep 2002** SPINNER Consortium (Regione Emilia-Romagna) Research Fellowship for Technology Transfer through a grant to the BioDec project.

Awards

- **Sep 2025** Included in the list of top-cited scientists by John P.A. Ioannidis - Updated science-wide author databases of standardized citation indicators. DOI: 10.17632/btchxk-tzyw.8 (https://elsevier.digitalcommonsdata.com/public-files/datasets/btchxktzyw/files/1b1cbb0c-9ec9-4482-ac6d-cc9d5586bd3f/file_downloaded)
- **Sep 2024** Included in the list of top-cited scientists by John P.A. Ioannidis - Updated science-wide author databases of standardized citation indicators. DOI:10.17632/btchxktzyw.7 (https://elsevier.digitalcommonsdata.com/public-files/datasets/btchxktzyw/files/5c0c334e-53d9-421b-b344-3ced6fc8d857/file_downloaded)
- **Oct 2023** Included in the list of top-cited scientists by John P.A. Ioannidis - Updated science-wide author databases of standardized citation indicators. DOI:10.17632/btchxktzyw.6 (https://elsevier.digitalcommonsdata.com/public-files/datasets/btchxktzyw/files/7bad01c2-63e8-4ed2-94f2-4af4b3b0a4fe/file_downloaded)
- **Mar 2023** National Scientific Habilitation (ASN) for Full Professor in Applied Physics (FIS/07).
- **Oct 2022** Included in the list of top-cited scientists by John P.A. Ioannidis - Updated science-wide author databases of standardized citation indicators. DOI:10.17632/btchxktzyw.4 (https://elsevier.digitalcommonsdata.com/public-files/datasets/btchxktzyw/files/dbefaa9c-0022-46dc-95bc-9e8f0ec820f7/file_downloaded)
- **Sep 2018** National Scientific Habilitation (ASN) for Full Professor in Biochemistry (BIO/10)
- **Apr 2017** National Scientific Habilitation (ASN) for Associate Professor in Molecular Biology (BIO/11)
- **Mar 2017** National Scientific Habilitation (ASN) for Associate Professor in Biochemistry (BIO/10)
- **Apr 2016** Direct recruitment from abroad (*Chiamata diretta*) - Senior Assistant Professor (RTD B) at the University of Bologna co-funded by the Italian Ministry of the University

and Research (MIUR)

Visiting

- **Jun 2012 – Jul 2012** Prof. Frederic Rousseau and Prof. Joost Schymkowitz, VIB Switch Laboratory, KU Leuven, Leuven (Belgium)
- **May 2009** Prof. Francisco Melo group. Department of Molecular Genetics and Microbiology. Pontificia Universidad Catolica de Chile, Santiago de Chile (Chile).
- **Jul 2008 - Aug 2008** Prof. Andrej Sali group, Departments of Biopharmaceutical Sciences and Pharmaceutical Chemistry, University of California at San Francisco (UCSF), San Francisco (CA).
- **Aug 2005 - Nov 2005** Prof. Jeffrey Skolnick group. Center of Excellence in Bioinformatics University of New York at Buffalo, Buffalo (NY)

Teaching activity

At the University of Bologna, I am coordinator of the International Master in Bioinformatics (since May 2022) and teacher of the second module of *Laboratory of Bioinformatics 1* and *Elements of Biophysics*. In the Bachelor in Genomics, I was teacher of the course *Proteomes, Interactomes and Biological Networks*. I am also member of the committee of the PhD program in Data Science and Computation. In the PhD program I am co-organizer of the course *Elements of Programming Languages* with Sergio Decherchi. I am also responsible for the ERASMUS exchange program with 7 foreign universities. The teaching material of my courses is available at `https://biofold.org/training.html`

Previously, from 2015 to 2016, at the University of Düsseldorf I taught two modules as part of the master courses *Introduction to Molecular Systems Biotechnology* and *Biological Networks*. From 2014 to 2015 at the University of Alabama at Birmingham, in collaboration with Dr. Malay Basu, I was teacher of two courses in the Graduate Biomedical Science program at UAB and organizer of the CB2 (Computational Biology and Bioinformatics) Journal Club at UAB. The list of courses given during last few years is reported below.

- **2023-2025** Elements of Programming Languages, PhD in Data Science and Computation (15 hours – Total 30 hours). University of Bologna.
- **2022-2025** Elements of Biophysics, International Master in Bioinformatics (32 hours/year). University of Bologna.
- **2025-2026** Bioinformatics for Systems and Synthetic Biology, International Master in Bioinformatics (32 hours/year). University of Bologna.
- **2016-2026** Module 2: Laboratory of Bioinformatics 1, International Master in Bioinformatics (60 hours/year). University of Bologna.
- **2019-2021** Proteomes, Interactomes and Biological Networks, Bachelor in Genomics (62 hours/year). University of Bologna.
- **2015-2016** Module: Introduction to Computational Biology and Bioinformatics. Introduction to Molecular Systems Biotechnology course (M4453). Molecular Systems Biotechnology Master (20 hours). University of Düsseldorf.
  Module: Introduction to Protein-Protein Interaction Network. Biological Networks course (M4424).
- **2014-2015** GBSC 703-01E Computational Biology and Bioinformatics, Graduate Biomedical Science Program (40 hours). University of Alabama at Birmingham (USA).
- **2014-2015** GBSC 703-01A - Introduction to Scientific Computing, Graduate Biomedical Science Program (48 hours). University of Alabama at Birmingham (USA).
- **2013-2014** GBSC 703-01A - Introduction to Scientific Computing, Graduate Biomedical Science Program (30 hours). University of Alabama at Birmingham (USA).

I was also tutor for the following courses held by Prof. Rita Casadio at the University of Bologna (Italy):

- **2004-2005** Bioinformatics - Degree in Biotechnology (30 hours)
- **2003-2004** Laboratory of Biophysics II - Degree in Biotechnology
  Models for Biological Systems - Degree in Biotechnology (90 hours total)
- **2002-2003** Structural Biochemistry - Degree in Biotechnology (70 hours)
- **2001-2002** Laboratory of Biophysics II - Degree in Biotechnology (25 hours)

Mentoring activity

At the University of Bologna, I am member of the committee of the PhD program in Data Science and Computation. Currently, I am supervisor of Chiara Rivi, Elisa Bonanni, and Carlos Arturo Enriquez Sandoval PhD students in the program of Data Science and Computation, Andrea Zauli Research Assistant at the Department of Pharmacy and Biotechnology.

I was supervisor of the research activity of Marco Antegini postdoc (May 2024 – May 2026) and Andrea Cicconardi, research assistant (Mar 2001 – Oct 2022). I co-mentored the research activity Jaume Sastre Tomàs, PhD student at the Department of Mathematics and Informatics, University of Balearic Islands (Spain).

In the past, I directed the master thesis of the following students:

- Marina Mariano Title: Prostate cancer detection using biparametric magnetic resonance imaging. Bologna: September 2026
- Elisa Bonanni Title: Characterization of mutational landscapes in leukemia in the context of clonal hematopoiesis. Bologna: 25 September 2025
- Chiara Rivi Title: Functional characterization and network analysis of genes associated with neurodevelopmental disorders. Bologna: 26 September 2024
- Lorenzo Lo Cascio Title: Development of machine learning methods for predicting the impact of protein mutants on the stability of protein complexes and their implication on the insurgence of human disorders Bologna 29 February 2024
- Cecilia Foglini Title: Predicting the effect of mutations on protein-protein interactions International Master in Bioinformatics at the University of Bologna (Italy). Bologna 29 September 2022
- Riccardo Ottalevi Title: Predicting the effect of single point mutations on protein-protein interactions International Master in Bioinformatics at the University of Bologna (Italy). Bologna 28 February 2022
- Tommaso Mosca Title: Impact of protein modeling on the prediction of protein stability change upon mutation. International Master in Bioinformatics at the University of Bologna (Italy). Bologna 28 February 2022
- Lucrezia Valeriani Title: Predicting the functional impact of single nucleotide variants in rare disease associated genes. International Degree in Genomics at the University of Bologna (Italy). Bologna 11 September 2020
- Alessandro Vinceti Title: Development of new tools for predicting the impact of genetic variants in the cancer genome. International Master in Bioinformatics at the University of Bologna (Italy). Bologna, 25 September 2019.
- Emina Merdan Title: Characterizing the impact of mutations at functional and network levels in Lung Adenocarcinoma. International Master in Bioinformatics at the University of Bologna (Italy). Bologna, 26 September 2018.
- Oronzo Tassiello Title: Computational methods for scoring genomes of Lung Adenocarcinoma International Master in Bioinformatics at the University of Bologna (Italy). Bologna, 9 March 2018.

- Luigi Chiricosta Title: Detecting cancer causing genes and variants in Colon Adenocarcinoma. International Master in Bioinformatics at the University of Bologna (Italy). Bologna, 26 September 2017.

At the University of Alabama at Birmingham, I was supervising the research activity of one postdoc (Dr. Rui Tian) and one master student (Shivani Viradia).

Previously, I collaborated with Prof. Rita Casadio, Dr Mario Compiani and Dr Marc A. Marti-Renom to mentor the research activity of the following students:

- Alberto Stizza BS thesis in Physical Sciences, Catholic University of Brescia (Italy)
- Daniela Danesi BS thesis in Physical Sciences, Catholic University of Brescia (Italy)
- Maria Procopio BS thesis in Physical Sciences, University of Bologna (Italy)
- Remo Calabrese PhD thesis in Biotechnology, University of Bologna (Italy)
- Giulia Gentile MS thesis in Bioinformatics, CRS4 Bioinformatics Laboratory Cagliari (Italy)
- Stefania Bosi PhD student, University La Sapienza, Roma (Italy)
- Miquel Oltra Sastre Master thesis, Open University of Catalonia (Spain)

Reviewer and editorial activities

Until 2022 I was faculty member of the Faculty Opinions in the section of Bioinformatics, Biomedical Informatics & Computational Biology / Translational Bioinformatics. I am reviewer for the following journals: Nature Communications, Bioinformatics, Briefings in Bioinformatics, Nucleic Acids Research, The American Journal of Human Genetics, PLOS Computational Biology, Scientific Reports, Cancer Research, Oncotarget, BMC Bioinformatics, PLOS ONE, BMC Genomics, Proteins, Human Mutation, Human Genetics, Human Genomics, Amino Acids, BMC Structural Biology, Database, Current Bioinformatics, Current Protein and Peptide Science, Journal of Bioinformatics and Computational Biology, Neurocomputing, Information Fusion, IEEE/ACM Transactions on Computational Biology and Bioinformatics. I was reviewer of projects for the Medical Research Council of the United Kingdom and for the Austrian Academy of Sciences.

I am also member of the editor board of the following journals:

- Scientific Reports (Computational Science)
- Frontiers in Bioinformatics (Protein Bioinformatics section)
- Frontiers in Molecular Biosciences (Molecular Diagnostics and Therapeutics section)
- Computation MDPI (Computational Biology section)

Selection and evaluation committees

- **2023** Evaluation Committee for PhD in Biomedical Sciences and Oncology at the University of Torino (Italy)
- **2022** Committee RTD A (Junior Assistant Professor) University of Bologna, D.D. 6352, 11/10/2022.
  Committee RTD B (Senior Assistant Professor) University "La Sapienza" Roma, D.R. 3227/2021, 02/12/2021
- **2020** Evaluation Committee PhD in Computer Sciences at the University College Dublin (Ireland)
  Candidate: Mirko Torrisi
- **2020** Evaluation Committee PhD in Biomedical Sciences at the University of Padova (Italy)
  Candidates: Antonella Falconieri, Federica Quaglia and Andras Hatos

- **2018** Evaluation of Marco Carraro PhD in Biomedical Sciences at the University of Padova (Italy)

Other scientific activities

- International Society of Computational Biology (ISCB)
  - 2011-present: co-chair in the organization of the Personal Genomics session at the Pacific Symposium of Biocomputing (PSB) 2011 and with Yana Bromberg, Hannah Carter and Antonio Rausell, I organized 15 editions of the VarI-COSI meeting (formerly SNP-SIG) in Vienna (Austria), Long Beach, (California), Berlin (Germany), Boston (Massachusetts), Dublin (Ireland), Orlando (USA), Prague (Czech Republic), Chicago (USA), Basel (Switzerland), Madison (Wisconsin), Lyon (France), Montreal (Canada), Washington (DC).
  - 2011-2022: co-editor of 6 special issues on BMC Genomics which published the selected works presented at the VarI-SIG meetings. I was also co-Editor of a special issue entitled "*VarI-COSI 2022: Identification and Annotation of Genetic Variants in the Context of Structure, Function and Disease*" published on the International Journal of Molecular Sciences. More information about the VarI-COSI meeting is available at https://varicosi.biofold.org/.
  - 2013: member of the Proceedings Papers Committee for the ISMB/ECCB Conference.
  - 2015: member of the Late Breaking Research Committee for the ISMB/ECCB Conference. In 2016 I was co-chair of the Disease track for the ISCB Latin American Conference in Buenos Aires (Argentina).
  - 2020: member of the proceeding committee of the ISMB Conference for the "Genomic Variation Analysis" track.
- CAGI -Critical Assessment of Genome Interpretation
  - 2010: member of the Data Committee.
  - 2018: organizer of the Frataxin challenge
  - 2021: organizer of the Calmodulin and MAPKs challenges.
  - 2025: assessor for the BARD1 and the ATP7B challenges.
- ELIXIR Member
  - 2026: Speaker of the Workshops *Bringing health data standards across Europe: From OMOP to GA4GH* and *Connecting health data pan-Europe: how can we help each other?*. ELIXIR AllHands Meeting 2026. 8-10 June 2026, Lyon (France).
  - 2025-Present: co-lead of the ELIXIR Rare Diseases Community
  - 2025: organization of the Rare Disease F2F Meeting, 15-16 May 2025, Bologna (Italy).
  - 2022-2023: co-coordinator of the Service Bundle 1 (https://amp4rd.github.io) of the Rare Disease Community
  - 2022-2023: ELIXIR AllHands Meetings 2023: Co-organizer of the Workshop *Enhancing machine learning application for Rare Disease*, ELIXIR Allhands Meeting, 5-8 June 2023, Dublin Ireland.
  - 2023: Co-organizer of the workshop *ELIXIR-IT Rare Disease Community*, 28-29 November 2023, Roma (Italy).
  - 2022: Co-organizer of the workshops *Identification, collection and review of datasets for the assessment and development of supervised machine learning methods for biomedical applications*, ELIXIR AllHands, 7-10 June 2022, Amsterdam (Netherlands)
- SIB Member
  - 2026: Co-organizer of the meeting of the SIB Computational and Systems Biology Group, 16 June 2026, Bologna (Italy).
  - 2024-present: Coordinator of the Computational and Systems Biology group of the Italian Society for Biochemistry and Molecular Biology (SIB).

– 2024: Chair of the speed talk session. The 48th FEBS Congress, 29 June - 3 July 2024, Milan (Italy).

Research Interests

- Analysis and interpretation of human genome.
- Genome interpretation and prediction of disease-related protein mutations.
- Machine learning approaches in molecular biology.
- Protein-protein interactions.
- RNA structure comparison and prediction.
- Protein folding kinetics.
- Prediction of protein stability changes upon mutation.
- Protein structural prediction by threading methods and building by homology.
- Molecular dynamics of protein systems.

Developed Web Servers, Tools and Databases

- **ContrastRank:** probabilistic method for cancer gene prioritization and cancer sample classification.
  WEB: http://snps.biofold.org/contrastrank
- **DrCancer:** predictor of cancer causing non-synonymous single nucleotide polymorphisms.
  WEB: http://snps.biofold.org/drcancer
- **DDGun:** untrained method for predicting protein stability upon single and multiple site mutations .
  WEB: http://folding.biofold.org/ddgun
- **Fido-SNP:** predicts the impact of genetic variants in coding and non-coding regions of the dog genome.
  WEB: http://snps.biofold.org/fido-snp
- **I-Mutant1.0:** Neural Network based method to predict the sign of free energy change of proteins upon single point mutation.
  WEB: http://gpcr2.biocomp.unibo.it/cgi/predictors/I-Mutant/I-Mutant.cgi
- **I-Mutant2.0:** Support Vector Machine based method to predict the sign and the value of free energy change of proteins upon single point mutation.
  WEB: http://folding.biofold.org/i-mutant
- **is-pero:** Machine learning method for predicting the localisation of peroxisomal proteins.
  WEB: https://structure.biofold.org/is-pero/
- **K-Fold:** Support Vector Machine based method to predict the mechanism and rate of protein folding kinetics.
  WEB: http://folding.biofold.org/k-fold
- **K-Pro:** Database of experimental data on the kinetics of protein and mutants.
  WEB: http://folding.biofold.org/k-pro
- **Meta-SNP:** Meta-predictor of disease-causing variants that uses PANTHER, PhD-SNP, SIFT and SNAP.
  WEB: http://snps.biofold.org/meta-snp
- **Omidios:** Omidios, a database of pre-calculated likely impact of a Single Nucleotide Polymorphism in the human genome.
  WEB: http://sgt.cnag.cat/services/Omidios/
- **PhD-SNP:** Support Vector Machine based Method to discriminate between disease-related and neutral mutations in proteins.
  WEB: http://snps.biofold.org/phd-snp
- **PhD-SNP^{g}:** A gradient boosting-based method for predicting the impact of genetic variants in coding and non-coding regions.

WEB: http://snps.biofold.org/phd-snpg
- **SARA:** a tool for Structural Alignment of Ribonucleic Acids.
WEB: http://structure.biofold.org/sara
- **SARA-Coffee:** tool for RNA multiple structural alignment obtained merging SARA and T-Coffee.
WEB: http://www.tcoffee.org/Projects/saracoffee/
- **ThermoScan:** tool for scanning biomedical publications to retrieve protein thermodynamic data.
WEB: https://folding.biofold.org/thermoscan
- **WebRASP:** statistical potential for scoring the quality of RNA three-dimensional structure.
WEB: http://melolab.org/webrasp
- **WS-SNPs&GO:** predictor of human disease related mutations in proteins with functional annotation.
WEB: http://snps.biofold.org/snps-and-go

International conferences meetings and schools

- The European Paediatric Transnational Research Infrastructure (EPTRI) Scientific Meeting, Warsaw (Poland) 28-29 July 2026.
- ELIXIR Human Copy Number Variants (hCNV) Community F2F Meeting, Paris (France) 7-8 July 2026.
- ELIXIR AllHands Meeting, Lyon (France), 8-10 June, 2026.
- ELIXIR Federated Human Data (FHD) Community F2F Meeting, Geneva (Switzerland) 10-11 November 2025.
- XXXIII Intelligent Systems for Molecular Biology meeting (ISMB), Liverpool (UK), 20-24 July, 2025.
- ELIXIR AllHands Meeting, Thessaloniki (Greece), 2-5 June, 2025.
- ESHG Conference 2025. Milan, Italy. May 24–27, 2025.
- ELIXIR AllHands Meeting 2023, Dublin (Ireland), 5-8 June 2023.
- XXI European Conference on Computational Biology (ECCB), Sitges (Spain), 18-21 September 2022.
- XXX Intelligent Systems for Molecular Biology meeting (ISMB), Madison, WI (USA), 10-14 July, 2022.
- VI Critical Assessment of Genome Interpretation. Berkeley CA (USA) 14-16 May, 2022.
- ELIXIR AllHands meeting 2022, Amsterdam (Netherlands), 7-10 June 2022.
- Moscow Conference on Computational Molecular Biology (MCCMB), Moscow (Russian Federation), 30 July - 2 August 2021.
- XXIX Intelligent Systems for Molecular Biology meeting and XX European Conference on Computational Biology (ECCB). Virtual Conference, 25-30 July 2021.
- XXVIII Intelligent Systems for Molecular Biology meeting (ISMB). Virtual Conference, 13-16 July 2020.
- Bologna Winter School 2020: *What can we learn from protein structure?* Bologna (Italy), 17-21 February 2020.
- Bologna Winter School 2019: *Data Science for Bioinformatics.* Bologna (Italy), 18-22 February 2019.
- XXVI Intelligent Systems for Molecular Biology meeting (ISMB), Chicago, (USA), 6-10 July 2018.
- V Critical Assessment of Genome Interpretation (CAGI). Chicago, (USA), 5-7 July 2018.
- The molecular basis of diseases: Can we infer phenotypes from protein variant analysis? FEBS Advanced Course. 23-25 May 2018, Bologna, Italy
- Bologna Winter School 2022: *Big Data and Bioinformatics.* Bologna (Italy), 12-16 Febru-

ary 2022.

- XXV Intelligent Systems for Molecular Biology meeting (ISMB) and XVI European Conference on Computational Biology (ECCB), Prague, (Czech Republic), 21-25 July 2017
- Bologna Winter School 2017: *Revisiting Bioinformatics Foundations.* Bologna (Italy), 13-17 February 2017.
- XIV Intelligent Systems for Molecular Biology meeting (ISMB), Orlando, FL (USA), 8-12 July 2016
- Bologna Winter School 2016: *In Silico Markers for Precision Medicine.* Bologna (Italy) 22-26 February 2016.
- EMBL Conference on Cancer Genomics, Heidelberg (Germany), 1-4 November 2015.
- XXIII Intelligent Systems for Molecular Biology meeting (ISMB) and XIV European Conference on Computational Biology (ECCB), Dublin (Ireland), 12-14 July 2015
- UAB NHGRI IV Short Course on Next-Generation Sequencing; Technology and Statistical Methods. Birmingham (AL), 15-18 December 2014.
- XIII European Conference on Computational Biology (ECCB), Strasbourg (France), 7-10 September 2014.
- XXII Intelligent Systems for Molecular Biology meeting (ISMB), Boston, Massachusetts (USA), 13-15 July 2014.
- VarI-SIG meeting. Identification and annotation of genetic variants in the context of structure, function, and disease. Boston, Massachusetts (USA), 12 July 2014.
- XXI Intelligent Systems for Molecular Biology meeting (ISMB) and XII European Conference on Computational Biology (ECCB), Berlin (Germany), 21-23 July 2013
- SNP-SIG meeting. Identification and annotation of SNPs in the context of structure, function, and disease. Berlin (Germany), 19 July 2013
- Critical Assessment of Genome Interpretation (CAGI). Berlin (Germany), 17-18 July 2013.
- ESHG Course in Next Generation Sequencing, Bertinoro di Romagna (Italy), 17-20 May 2013
- Summit on Translational Bioinformatics (TBI), San Francisco, California (USA), 18-20 March 2013.
- XX Intelligent Systems for Molecular Biology meeting (ISMB), Long Beach, California (USA), 15-17 July 2012.
- SNP-SIG meeting. Identification and annotation of SNPs in the context of structure, function, and disease. Long Beach, California (USA), 14 July 2012.
- Bologna Winter School 2012 *Proteins and their variants: structure and function prediction.* Bologna (Italy) 13-17 February 2012.
- XIX Intelligent Systems for Molecular Biology meeting (ISMB) and X European Conference on Computational Biology (ECCB), Vienna (Austria), 17-19 July 2011.
- SNP-SIG meeting. Identification and annotation of SNPs in the context of structure, function, and disease. Vienna (Austria), 15 July 2011.
- EMBO Young Scientist Forum. International Institute of Molecular and Cell Biology (IIMCB), Warsaw (Poland), June 30$^{\text{th}}$ – July 1$^{\text{st}}$ 2011.
- Pacific Symposium on Biocomputing (PSB) 2011. Big Islands, Hawaii January 3-7 2011.
- Critical Assessment of Genome Interpretation (CAGI). University of California at Berkeley. Berkeley, California (USA), 10 December 2010.
- Biomedical Computation at Stanford (BCATS). Stanford University. Palo Alto, California (USA), 6 November 2010.
- Exploring the functional consequences of genomic variation (HGVS meeting), Washington DC (USA), 2 November 2010
- II Workshop on Annotation, Interpretation and Management of Mutations (AIMM) and IX European Conference on Computational Biology (ECCB), Ghent (Belgium), 26-29 September 2010.

- 4th Comprehensive Cancer Research Training Program (CCRTP) at Stanford University, Palo Alto California (USA), 13-17 September 2010.
- 9th International Conference on Computational Systems Bioinformatics (CSB). Stanford, Palo Alto, California (USA), 16-18 August 2010.
- XVIII Intelligent Systems for Molecular Biology meeting (ISMB), Boston (USA), 11-13 July 2010.
- Biomedical Computation at Stanford (BCATS). Stanford University. Palo Alto, California (USA), 7 November, 2009.
- Lipari International Summer School on BioInformatics and Computational Biology. RNAs: structure, function and therapy. Lipari (ME) 13-20 June, 2009.
- VII European Conference on Computational Biology (ECCB), Cagliari (Italy), 22-26 September 2008.
- Workshop on Applications of Protein Models in Biomedical Research, University of California San Francisco (UCSF), San Francisco (CA) 11-12 July, 2008
- III Course on Molecular Evolution, Phylogenetics and Phylogenomics, Valencia (Spain) 12-16 May 2008.
- Non-Coding RNAs: Computational Challenges and Applications. Antalya (Turkey) 28-30 April 2008.
- XV Intelligent Systems for Molecular Biology meeting (ISMB) and VI European Conference on Computational Biology (ECCB), Vienna (Austria), 21-25 July 2007.
- ISMB 3DSig Satellite Meeting - Structural Bioinformatics and Computational Biophysics, Vienna (Austria), 19-20 July 2007.
- EMBO Workshop: Viral RNA: Structure Function and Targeting. EMBL Heidelberg (Germany) 5-7 March 2007.
- Bologna Winter School 2006. Applied Bioinformatics: The test case of Human Genome. Bologna (Italy), 13-17 February, 2006.
- Bologna Winter School 2005: *How Complex is Functional Genomics?* Bologna (Italy), 13-19 February 2005.
  XII Intelligent Systems for Molecular Biology (ISMB) and III European Conference on Computational Biology meeting (ECCB), Glasgow (Scotland) 31 July – 4 August 2004.
- Bologna Winter School 2004: *The State of the Art of Protein-Protein Interaction Networks. The role of the "in silico" approach*, Bologna (Italy) 8-14 February 2004.
- Meeting Galileo Project. Marseille (France) 27-28 June 2003.
- Bologna Winter School 2003: *Hot Topics in Structural Genomics* Bologna (Italy) 9-15 February 2003.
- Bologna Winter School 2002: *Predicting 3D Structure of Difficult Proteins.* Bologna (Italy) 3-9 February 2002.
- Bologna Winter School 2001: *In Silico Biomolecular Recognition.* Bologna (Italy) 4-10 February 2001.
- Bologna Summer School: *Biotechnology Protein Sequence Analysis in the Genomic Era.* Bologna (Italy) 10-16 October 1999.

National conferences meetings and schools

- ELIXIR IT AllHands Meeting, Bari 9-10 February 2026.
- 28th Congress of the Italian Society of Human Genetics (SIGU), Rimini, 23-25 September 2025.
- 63th Congress of the Italian Society of Biochemistry (SIB), Palermo 10-12 September 2025.
- 27th Congress of the Italian Society of Human Genetics (SIGU), Padova, 2-4 October 2025.
- 62th Congress of the Italian Society of Biochemistry (SIB), Firenze 7-9 September 2023. 7-9 September 2023)

- 60th Congress of the Italian Society of Biochemistry (SIB), Lecce 18-20 September 2019.
- Computational and Translational Methods for Cancer Genomics, Bologna 29 May, 2018.
- German Conference on Bioinformatics (GCB) 2015, Dortmund (Germany), 27-30 September 2015.
- UAB Comprehensive Cancer Center, 15^{th} Annual Research Retreat and Research Competition, Birmingham, Alabama (USA), October 29 2012.
- VIII Jornadas de Bioinformatica. Valencia (Spain), 13-15 February 2008.
- VI Meeting on Nucleic Acids and Nucleotides (RANN07), Valencia (Spain), 22-23 November 2007.
- VII Jornadas de Bioinformatica. Zaragoza (Spain), 20-22 November 2006.
- Bioinformatics Italian Society (BITS) Annual Meeting 2006. Bologna (Italy), 28-29 April 2006.
- Bioinformatics Italian Society (BITS) Annual Meeting 2004 Padova (Italy), 26-27 March 2004.
- Workshop Staminal Cells: *Properties and Perspectives.* Bressanone (Italy), 11-13 September 2003.
- XI National School of Biophysics: Biophysics of the Cell. Bressanone (Italy), 8-10 September 2003.
- IX National School of Biophysics: *Biophysics and Biomaterials.* Bressanone (Italy), 3-5 September 2001.
- XXXI National Congress of Physical Chemistry, Padova (Italy), 19-23 June 2001.

Invited talks

- **07 Aug 2006**: Centro de Investigacion Principe Felipe (CIPF), Valencia (Spain)
- **23 Oct 2007**: Centro Nacional de Investigaciones Oncológicas (CNIO), Madrid (Spain)
- **06 May 2008**: Département d'Informatique, Université Libre de Bruxelles (ULB), Bruxelles (Belgium)
- **23 Apr 2009**: Departament de Ciències Matemàtiques i Informàtica, Universitat de les Illes Balears (UIB), Palma de Mallorca (Spain)
- **27 May 2010**: Buck Institute, Novato (California, USA)
- **02 Jul 2010**: Luxembourg Centre for System Biomedicine, Luxemburg University, Luxembourg
- **20 Sep 2010**: Department of Genetics and Bioengineering, Yeditepe University, Istanbul (Turkey)
- **25 Jan 2011**: Department of Medicinal Chemistry and Molecular Pharmacology, Purdue University, Lafayette (Indiana, USA).
- **18 Feb 2011**: Lawrence Berkeley National Laboratory, Berkeley (California, USA).
- **21 Mar 2011**: Department of Computer Sciences, Wayne State University, Detroit (Michigan, USA).
- **06 Apr 2011**: J. Craig Venter Institute, San Diego (California, USA)
- **22 Apr 2011**: Department of Bioengineering, University of Texas at Dallas (Texas, USA)
- **17 May 2011**: Department of Pathology, University of Alabama at Birmingham (Alabama, USA)
- **26 May 2011**: Department of Computer Science, Université Pierre et Marie Curie, Paris (France)
- **16 Jun 2011**: Instituto Gulbenkian de Ciencia, Oeiras (Portugal)
- **29 Jun 2011**: International Institute of Molecular and Cell Biology, Warsaw (Poland)
- **04 Oct 2011**: Institut de Cancerologie Gustave Roussy, Villejuif (France)
- **06 Oct 2011**: Karlsruhe Institute of Technology, Karlsruhe (Germany)
- **12 Jun 2012**: Switch Lab, KU Leuven, Leuven (Belgium)
- **30 Apr 2013**: Biomedical Informatics Day, Adelaide (SA, Australia)

- **16 Jul 2013**: Institute for Medical and Human Genetics, Charité University, Berlin (Germany)
- **07 Oct 2013**: CCNR, Northeastern University, Boston (Massachusetts, USA)
- **10 Dec 2013**: Macromolecular Biochemistry Research Center (CRBM), CNRS, Montpellier (France)
- **14 Dec 2013**: Computational Biology Institute (IBC), CNRS, Montpellier (France)
- **12 Feb 2014**: Institute of Genetics and Molecular and Cell Biology (IGBMC), Strasbourg (France)
- **22 May 2014**: Montpellier Cancer Research Institute, University of Montpellier, Montpellier (France)
- **06 Aug 2014**: Izmir Biomedicine and Genome Center (IBG), Izmir (Turkey)
- **18 Sep 2014**: Technical University of Munich (TUM), Munich (Germany)
- **26 Sep 2014**: Pontificia Universidad Catolica de Chile, Santiago de Chile (Chile)
- **12 Sep 2016**: Adam Mickiewicz University (AMU), Poznan (Poland)
- **17 Jul 2019**: Department of Computer Science, University of Lisboa, Lisboa (Portugal)
- **02 Aug 2021**: Moscow Conference on Computational Molecular Biology, Moscow (Russian Federation)
- **20 Jul 2022**: Genomic Medicine Institute, Cleveland Clinic, Cleveland OH (USA)
- **7 Dec 2025**: Assessment ATP7B BARD1 Missense Challenges. CAGI7 Conference. 6-8 December 2025, Berkeley, (California), USA.
- **07 Jul 2026**: ELIXIR human CNV Face-to-Face Meeting, Paris (France).
- **28 Jul 2026**: EPTRI Scientific Meeting, Warsaw (Poland).

Invited lectures

- **01 Sep 2011**: Statistics and Genomics Seminar, University of California, Berkeley (California, USA)
- **17 Feb 2012**: Bologna Winter School 2012, Bologna (Italy)
- **08 Feb 2013**: Anatomic Pathology Didactic Conference. University of Alabama at Birmingham (Alabama, USA)
- **15 Feb 2013**: Genetics and Genomics Seminar Series. University of Alabama at Birmingham (Alabama, USA)
- **04 Mar 2013**: GBM-722 Bioinformatics Course. University of Alabama at Birmingham (Alabama, USA)
- **07 Mar 2013**: Biotechnology Professional Master. University of Alabama at Birmingham (Alabama, USA)
- **22 Apr 2013**: GBS Structural Biology Course. University of Alabama at Birmingham (Alabama, USA)
- **14 May 2013**: Laboratory Medicine Seminar. University of Alabama at Birmingham (Alabama, USA)
- **17 May 2013**: European School of Genetic Medicine. European School of Genetic Medicine, Bertinoro (Forli', Italy)
- **02 Feb 2014**: GBM-722 Bioinformatics Course. University of Alabama at Birmingham (Alabama, USA)
- **06 Mar 2014**: Biotechnology Professional Master. University of Alabama at Birmingham (Alabama, USA)
- **03 Jul 2014**: GBS-758 New Perspectives in Cardiovascular Biology. University of Alabama at Birmingham (Alabama, USA)
- **08 Oct 2014**: Biotechnology Professional Master. University of Alabama at Birmingham (Alabama, USA)
- **18 Dec 2014**: UAB NHGRI IV Short Course on Next-Generation Sequencing; Technology and Statistical Methods. University of Alabama at Birmingham (Alabama, USA)

- **25/26 Feb 2016**: Bologna Winter School 2016, University of Bologna (Italy).
- **09 Jun 2017**: PhD Program in Biochemistry, "La Sapienza" University, Roma (Italy)
- **06 Sep 2017**: Special course on NGS data analysis. CRO National Cancer Institute, Aviano (Italy)
- **17 Jan 2019**: Winter School University of Verona, Canazei (Italy)
- **7 Jul 2022**: Pathology and Laboratory Medicine Grand Rounds, University of Kansas Medical Center, (Kansas, USA)
- **16 May 2023**: PhD in Biomedical Sciences and Oncology seminar series, University of Torino (Italy)

Publications

I published 79 peer-reviewed manuscripts 65 of which research articles and 14 reviews in international journals with impact factor. I also published 19 between book chapters/proceedings (11) and meeting reports (8). Using Google Scholar my papers received ∼10,700 citations corresponding to an h-index of 39. According to Scopus, my papers received ∼7,500 citations corresponding to an h-index of 33. Using Web of Science my articles received more than 6,600 citations corresponding to h-index of 32.

Google Scholar: https://bit.ly/ecapriotti-scholar
Scopus: https://bit.ly/ecapriotti-scopus
ORCID: https://bit.ly/ecapriotti-orcid
WOS: https://bit.ly/ecapriotti-wos
ResearchGate: https://bit.ly/ecapriotti-researchgate

Papers on international journals with impact factor

1. **Capriotti E**, Fariselli P, Rossi I, Casadio R (2004). A Shannon entropy-based filter detects high-quality profile-profile alignments in searches for remote homologues. **Proteins** 54:351-360.
2. Compiani M, **Capriotti E**, Casadio R (2004). The dynamics of the minimally frustrated helices determine the hierarchical folding of small helical proteins. **Phys Rev E**. 65:051905-8.
3. **Capriotti E**, Fariselli P, Casadio R (2004). A neural network-based method for predicting protein stability changes upon single point mutations. **Bioinformatics.** 20 (Suppl 1):I63-I68.
4. Stizza A, **Capriotti E**, Compiani M (2005). A minimal model of three-state folding dynamics of helical proteins. **Journal of Physical Chemistry B,** 109: 4215-4226.
5. **Capriotti E**, Fariselli P, Casadio R (2005). I-Mutant2.0: predicting stability changes upon mutation from the protein sequence or structure. **Nucleic Acids Research**, 33 Web Server Issue: W306-W310.
6. **Capriotti E**, Fariselli P, Calabrese R, Casadio R (2005). Predicting protein stability changes from sequences using support vector machines. **Bioinformatics,** 21 Suppl 2:ii54-ii58.
7. **Capriotti E**, Compiani M (2006). Diffusion-Collision of Foldons Elucidates the Kinetic Effects of Point Mutations and Suggests Control Strategies of the Folding Process of Helical Proteins. Proteins, 64: 198-209.
8. Grandi F, Sandal M, Guarguaglini G, **Capriotti E**, Casadio R, Samorì B (2006). Hierarchical mechanochemical switches in Angiostatin. **ChemBiochem,** 7; 1774-1782.
9. **Capriotti E**, Calabrese R, Casadio R. (2006). Predicting the insurgence of human genetic diseases associated to single point protein mutations with Support Vector Machines and evolutionary information. **Bioinformatics,** 22; 2729-2734.

10. **Capriotti E***, Casadio R (2007). K-Fold: a tool for the prediction of the protein folding kinetic order and rate. **Bioinformatics,** 23; 385-386.
11. **Capriotti E**, Arbiza L, Casadio R, Dopazo J, Dopazo H, Marti-Renom MA (2008). Use of estimated evolutionary strength at the codon level improves the prediction of disease related protein mutations in human. **Human Mutation**, 29; 198-204.
12. **Capriotti E**, Marti-Renom M. (2008). RNA structure alignment by a unit-vector approach. **Bioinformatics**, 24; i112-i116.
13. **Capriotti E**, Fariselli P, Rossi I, Casadio R. (2008). A three-state prediction of single point mutations on protein stability changes. **BMC Bioinformatics**. 9 (Suppl 2); S6.
14. Calabrese R, **Capriotti E**, Fariselli P, Martelli PL, Casadio R (2009) Functional annotations improve the predictive score of human disease-related mutations in proteins. **Human Mutation**, 30; 1237-1244.
15. **Capriotti E**, Marti-Renom M (2009) SARA: a server for function annotation of RNA structures. **Nucleic Acids Research.** 37 (Web Server issue); W260-W265.
16. **Capriotti E**, Marti-Renom MA. (2010). Quantifying the relationship between sequence and three-dimensional structure conservation in RNA. BMC Bioinformatics. 11; 322.
17. Baù D, Sanyal A, Lajoie BR, **Capriotti E**, Byron M, Lawrence JB, Dekker J, Marti-Renom MA (2011). The three-dimensional folding of the -globin gene domain reveals formation of chromatin globules. Nature Structural & Molecular Biology. 18; 107-114.
18. **Capriotti E**, Norambuena T, Marti-Renom MA, Melo F. (2011). All-atom knowledge-based potential for RNA structure prediction and assessment. Bioinformatics. 27; 1086-1093.
19. **Capriotti E***, Altman RB. (2011). Improving the prediction of disease-related variants using protein three-dimensional structure. BMC Bioinformatics. 12 (Suppl 4); S3.
20. **Capriotti E***, Altman RB. (2011). A new disease-specific machine learning approach for the prediction of cancer-causing missense variants. Genomics. 98; 310-317.
21. Dewey FE, Chen R, Cordero SP, Ormond KE, Caleshu C, Karczewski KJ, Whirl-Carrillo M, Wheeler MT, Dudley JT, Byrnes JK, Cornejo OE, Knowles JW, Woon M, Sangkuhl K, Gong L, Thorn CF, Hebert JM, **Capriotti E**, David SP, Pavlovic A, West A, Thakuria JV, Ball MP, Zaranek AW, Rehm HL, Church GM, West JS, Bustamante CD, Snyder M, Altman RB, Klein TE, Butte AJ, Ashley EA. (2011). Phased whole-genome genetic risk in a family quartet using a major allele reference sequence. PLOS Genetics 7; e1002280.
22. Kemena C, Bussotti G, **Capriotti E**, Marti-Renom MA, Cedric Notredame C. (2013). Using tertiary structure for the computation of highly accurate multiple RNA alignments with the SARA-Coffee package. Bioinformatics. 29:1112-1119.
23. **Capriotti E***, Calabrese R, Fariselli P, Martelli PL, Altman RB, Casadio R*. (2013). WS-SNPs&GO: a web server for predicting the deleterious effect of human protein variations using functional annotation. BMC Genomics. 14 (Suppl 3): S6.
24. **Capriotti E***, Altman RB, Bromberg Y*. (2013) Collective judgment predicts disease-associated single nucleotide variants. BMC Genomics. 14 (Suppl 3): S2.
25. Norambuena T, Cares JF, **Capriotti E**, Melo F (2013). WebRASP: a server for computing energy scores to assess the accuracy and stability of RNA 3D structures. Bioinformatics. 29: 2649-2650.
26. Li B, Seligman C, Thusberg J, Miller JL, Auer J, Whirl-Carrillo M, **Capriotti E**, Klein TE, Mooney SD. (2014). In silico comparative characterization of pharmacogenomic missense variants. BMC Genomics. 15 (Suppl 4): S4.
27. Di Tommaso P, Bussotti G, Kemena C, **Capriotti E**, Chatzou M, Prieto Barja P, Notredame C. (2014). SARA-Coffee web server, a tool for the computation of RNA sequence and structure multiple alignments. Nucleic Acids Research. 42 (Web Server Issue): W356-W360.
28. Tian R, Basu MK,**Capriotti E***. (2014). ContrastRank: a new method for ranking putative cancer driver genes and classification of tumor samples. Bioinformatics. 30:

i572-i578.
29. Beerten J, Van Durme J, Gallardo R, **Capriotti E**, Serpell L, Schymkowitz J and Rousseau F (2015). WALTZ-DB: a benchmark database of amyloidogenic hexapep-tides. Bioinformatics. 31:1698-1700.
30. Khass M, Blackburn T, Burrows PD, Walter MR, **Capriotti E**, Schroeder HW Jr. (2016). VpreB serves as an invariant surrogate antigen for selecting immunoglobulin antigen-binding sites. Science Immunology. 1: DOI: 10.1126/sciimmunol.aaf6628.
31. **Capriotti E***, Martelli PL, Fariselli P, Casadio R. (2017). Blind prediction of deleterious amino acid variations with SNPs&GO. Human Mutation. DOI:10.1002/humu.23179
32. Carraro M, Minervini G, Giollo M, Bromberg Y, **Capriotti E**, Casadio R, Dunbrack R, Elefanti L, Fariselli P, Ferrari C, Gough J, Katsonis P, Leonardi E, Lichtarge O, Menin C, Martelli PL, Niroula A, Pal LR, Repo S, Scaini MC, Vihinen M, Wei Q, Xu Q, Yang Y, Yin Y, Zaucha J, Zhao H, Zhou Y, Brenner SE, Moult J, Tosatto SCE. (2017). Performance of in silico tools for the evaluation of p16INK4a (CDKN2A) variants in CAGI. Human Mutation. DOI:10.1002/humu.23235.
33. **Capriotti E***, Fariselli P. (2017). PhD-SNP^{g}: A webserver and lightweight tool for scoring single nucleotide variants. Nucleic Acids Research. DOI:10.1093/nar/gkx369.
34. **Capriotti E***, Montanucci L, Profiti G, Rossi I, Giannuzzi D, Aresu L, Fariselli P*. (2019). Fido-SNP: The first webserver for scoring the impact of single nucleotide variants in the dog genome. Nucleic Acids Research. DOI: 10.1093/nar/gkz420
35. Savojardo C, Babbi G, Bovo S, **Capriotti E**, Martelli PL, Casadio R. (2019). Are machine learning based methods suited to address complex biological problems? Lessons from CAGI-5 challenges. Human Mutation. DOI: 10.1002/humu.23784.
36. Petrosino M, Pasquo A, Novak L, Toto A, Gianni S, Mantuano E, Veneziano L, Minicozzi V, Pastore A, Puglisi R, **Capriotti E**, Chiaraluce R, Consalvi V. (2019). Characterization of human frataxin missense variants in cancer tissues. Human Mutation. DOI: 10.1002/humu.23789.
37. Montanucci L, **Capriotti E***, Frank Y, Ben-Tal N, Fariselli P*. (2019). DDGun: an untrained method for the prediction of protein stability changes upon single and multiple point variations. BMC Bioinformatics. 20 (Suppl 14): 335.
38. Savojardo C, Petrosino M, Babbi G, Bovo S, Corbi-Verge C, Casadio R, Piero Fariselli P, Folkman L, Garg A, Karimi M, Katsonis P, Kim PM, Lichtarge O, Martelli PL, Pasquo A, Pal D, Shen Y, Strokach AV, Turina P, Zhou Y, Andreoletti G, Brenner S, Chiaraluce R, Consalvi V,**Capriotti E***. (2019). Evaluating the predictions of the protein stability change upon single amino acid substitutions for the FXN CAGI5 challenge. Human Mutation. DOI: 10.1002/humu.23843.
39. Kasak L, Bakolitsa C, Hu Z, Yu C, Rine J, Dimster-Denk DF, Pandey G, De Baets G, Bromberg Y, Cao C, **Capriotti E**, Casadio R, Van Durme J, Giollo M, Karchin R, Katsonis P, Leonardi E, Lichtarge O, Martelli PL, Masica D, Mooney SD, Olatubosun A, Pal LR, Radivojac P, Rousseau F, Savojardo C, Schymkowitz J, Thusberg J, Tosatto SCE, Vihinen M, Väliaho J, Repo S, Moult J, Brenner SE, Friedberg I. (2019). Assessing Computational Predictions of the Phenotypic Effect of Cystathionine-beta-Synthase Variants. Human Mutation. DOI: 10.1002/humu.23868.
40. Zhang J, Kinch LN, Cong Q, Katsonis P, Lichtarge O, Savojardo C, Babbi G, Martelli PL, **Capriotti E**, Casadio R, Garg A, Pal D, Weile J, Sun S, Verby M, Roth FP, Grishin NV. (2019) Assessing predictions on fitness effects of missense variants in calmodulin. Human Mutation. DOI: 10.1002/humu.23857.
41. Monzon AM, Carraro M, Chiricosta L, Reggian F, Han J, Ozturk K, Wang Y, Miller M, Bromberg Y, **Capriotti E**, Savojardo C, Babbi G, Martelli PL, Casadio R, Katsonis P, Lichtarge O, Carter H, Kousi M, Katsanis N, Andreoletti G, Moult J, Brenner SE, Ferrari C, Leonardi E, Tosatto SCE. (2019). Performance of computational methods for

the evaluation of Pericentriolar Material 1 missense variants in CAGI-5. Human Mutation. DOI: 10.1002/humu.23856.

42. Voskanian A, Katsonis P, Lichtarge O, Pejaver V, Radivojac P, Mooney SD, **Capriotti E**, Bromberg Y, Wang Y, Miller M, Martelli PL, Savojardo C, Babbi G, Casadio R, Cao Y, Sun Y, Shen Y, Garg A, Pal D, Yu Y, Huff CD, Tavtigian SV, Young E, Neuhausen SL, Ziv E, Pal LR, Andreoletti G, Brenner S, Moult J, Kann MG. (2019). Assessing the performance of in-silico methods for predicting the pathogenicity of variants in the gene CHEK2, among Hispanic females with breast cancer. Human Mutation. DOI: 10.1002/humu.23849.
43. McInnes G, Daneshjou R, Katsonis P, Lichtarge O, Srinivasan RG, Rana S, Radivojac P, Mooney SD, Pagel KA, Stamboulian M, Jiang Y, **Capriotti E**, Wang Y, Bromberg Y, Bovo S, Savojardo C, Martelli PL, Casadio R, Pal LR, Moult J, Brenner S, Altman R. (2019). Predicting venous thromboembolism risk from exomes in the Critical Assessment of Genome Interpretation (CAGI) challenges. Human Mutation. DOI: 10.1002/humu.23825.
44. Benevenuta S, **Capriotti E***, Fariselli P* (2021). Calibrating variant-scoring methods for clinical decision making. Bioinformatics. DOI: 10.1093/bioinformatics/btaa943.
45. Birolo G, Benevenuta S, Fariselli P*, **Capriotti E***, Giorgio E, Sanavia T. (2021). Protein stability perturbation contributes to the loss of function in haploinsufficient genes. Frontiers in Molecular Biosciences. 8: 620793.
46. Turina P, Fariselli P*,**Capriotti E***. (2021). ThermoScan: Semi-automatic identification of protein stability data from PubMed. Frontiers in Molecular Biosciences. DOI:10.3389/fmolb.2021.620475.
47. Pancotti C, Benevenuta S, Repetto V, Birolo G, **Capriotti E**, Sanavia T, Fariselli P. (2021). A deep-learning sequence-based method to predict protein stability changes upon genetic variations. Genes. 12:911. DOI:10.3390/genes12060911.
48. Walsh I, Fishman D, Garcia-Gasulla D, Titma T, Pollastri G, ELIXIR Machine Learning Focus Group (Capriotti E, Casadio R Capella-Gutierrez S, Cirillo D, Del Conte A, Dimopoulos AC, Dominguez Del Angel V, Dopazo J, Fariselli P, Fernández JM , Huber F, Kreshuk A, Lenaerts T, Martelli PL, Navarro A, Ó Broin P, Piñero J, Piovesan D, Reczko M, Ronzano F, Satagopam V, Savojardo C, Spiwok V, Tangaro MA, Tartari G, Salgado D, Valencia A Zambelli F), Harrow J, Psomopoulos FE, Tosatto SCE (2021). DOME: recommendations for supervised machine learning validation in biology. Nature Methods. 18:1122-1127. DOI:10.1038/s41592-021-01205-4.
49. **Capriotti E***, Fariselli P* (2022). Evaluating the relevance of sequence conservation in the prediction of pathogenic missense variants. Human Genetics. 141:1649-1658. DOI:10.1007/s00439-021-02419-4.
50. Pancotti C, Benevenuta S, Birolo G, Repetto V, Alberini V, Sanavia T, **Capriotti E***, Fariselli P* (2022). Predicting protein stability changes upon single-point mutation: a thorough comparison of the available tools on a new dataset. Briefings in Bioinformatics. 23:bbab555. DOI:10.1093/bib/bbab555
51. Montanucci L[†], Capriotti E[†], Birolo G, Benevenuta S, Pancotti C, Lal D, Fariselli P* (2022). DDGun: an untrained predictor of protein stability changes upon amino acid variants. Nucleic Acids Research. 50(W1):W222-W227. DOI: 10.1093/nar/gkac325
52. Benevenuta S, Birolo G, Sanavia T, **Capriotti E**, Fariselli P. (2023) Challenges in predicting stabilizing variations: An exploration. Frontiers in Molecular Biosciences. 9:1075570. DOI: 10.3389/fmolb.2022.1075570.
53. **Capriotti E***, Fariselli P*. (2023). PhD-SNP^{g}: updating a webserver and lightweight tool for scoring nucleotide variants. Nucleic Acids Research. DOI:10.1093/nar/gkad455.
54. Rocha J*, Sastre S, Amengual-Cladera E, Hernandez-Rodriguez J, Asensio-Landa V, Heine-Suner D, **Capriotti E**, (2023). Identification of driver epistatic gene pairs combining germline and somatic mutations in cancer. International Journal of Molecular Sciences. 24:9323. DOI: 10.3390/ijms24119323. (IF 2021: 6.208, Q1)

55. Turina P, Fariselli P,**Capriotti E***. (2023). K-Pro: Kinetics data on proteins and mutants. Journal of Molecular Biology. DOI:10.1016/j.jmb.2023.168245.
56. Petrosino M, Novak L, Pasquo A, Turina P, **Capriotti E**, Minicozzi V, Consalvi V, Chiaraluce R. (2023). The complex impact of cancer-related missense mutations on the stability and on the biophysical and biochemical properties of MAPK1 and MAPK3 somatic variants. Human Genomics. 17:95. DOI:10.1186/s40246-023-00544-x.
57. Critical Assessment of Genome Interpretation Consortium. (2024). CAGI, the Critical Assessment of Genome Interpretation, establishes progress and prospects for computational genetic variant interpretation methods. Genome Biology. 17:95. DOI:10.1186/s13059-023-03113-6.
58. Turina P, Dal Cortivo G, Enriquez Sandoval CA, Alexov E, Ascher DB, Babbi G, Bakolitsa C, Casadio R, Fariselli P, Folkman L, Kamandula A, Katsonis P, Li C, Lichtarge O, Martelli PL, Panday SK, Pires DEV, Portelli S, Pucci F, Rodrigues CHM, Rooman M, Savojardo C, Schwersensky M, Shen T, Strokach AV, Sun Y, Woo J, Radivojac P, Brenner SE, Dell'Orco D, **Capriotti E***. (2024). Assessing the predicted impact of single amino acid substitutions in calmodulin for CAGI6 challenges. Human Genetics. DOI:10.1007/s00439-024-02720-y.
59. Abdelghani Attafi O, Clementel D, Kyritsis K, **Capriotti E**, Farrell G, Fragkouli SC, Castro LJ, Hatos A, Lenaerts T, Mazurenko S, Mozaffari S, Pradelli F, Ruch P, Savojardo C, Turina MP, Zambelli F, Piovesan D, Monzon AM, Psomopoulos F, Tosatto SCE. (2024). DOME Registry: Implementing community-wide recommendations for reporting supervised machine learning in biology. GigaScience. 13:giae094.
60. Zhang J, Kinch L, Katsonis P, Lichtarge O, Jagota M, Song YS, Sun Y, Shen Y, Kuru N, Dereli O, Adebali O, Alladin MA, Pal D, **Capriotti E**, Turina MP, Savojardo C, Martelli PL, Babbi G, Casadio R, Hu Z, Pucci F, Rooman M, Cia G, Tsishyn M, Strokach A, van Loggerenberg W, Roth FP, Radivojac P, Brenner SE, Cong Q, Grishin NV. (2024). Assessing predictions on fitness effects of missense variants in HMBS in CAGI6. Human Genetics. DOI:10.1007/s00439-024-02680-3
61. Trucchi E, Fuselli S, Perta N, Partergnani S, Danese A, Balestra D, **Capriotti E**, Di Muccio G, Di Marino D, Pinton P, Bertorelle. G. (2025). A fixed mutation in the respiratory complex I impairs mitochondrial bioenergetics in the endangered Apennine brown bear. Proc. Natl. Acad. Sci. U.S.A. 122: e2504409122. DOI:10.1073/pnas.2504409122.
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